

Smart Helmet System for Advanced Safety Using IoT

This project is an **IoT-based Smart Helmet System** that improves motorcycle rider safety by combining **helmet detection, alcohol detection, accident detection, GPS tracking, GSM emergency alerts, and real-time monitoring**. Unlike a traditional helmet, this smart helmet communicates with the motorcycle and emergency contacts to prevent accidents and provide immediate help when an accident occurs. The system uses an **Arduino Uno** as the main controller and integrates multiple sensors with IoT technologies.

What Problem Does This Project Solve?

The paper focuses on several major problems faced by two-wheeler riders:

1. Riders do not wear helmets.

Many riders ignore helmet rules, leading to serious head injuries during accidents.

2. Drunk Driving.

Riding after consuming alcohol significantly increases accident risk.

3. Delayed Emergency Response.

If an accident happens in a remote area, medical assistance may arrive too late.

4. No Automatic Accident Detection.

Traditional helmets cannot detect accidents or notify emergency services.

5. No Live Tracking.

Family members and emergency responders cannot easily determine the rider's location after an accident.

6. Lack of Real-Time Monitoring.

There is no system to continuously monitor rider safety while traveling.

The proposed smart helmet addresses all of these issues using sensors, wireless communication, GPS, and GSM technologies.

Main Idea of the Project

The project transforms a normal motorcycle helmet into an **intelligent safety device**.

The helmet continuously checks:

- Whether the rider is wearing the helmet.
- Whether the rider has consumed alcohol.
- Whether the rider has met with an accident.

If all safety conditions are satisfied, the bike ignition is enabled. During the ride, the system continuously monitors for accidents. If an accident occurs, it automatically sends the rider's GPS location to emergency contacts through GSM while displaying system status on an LCD and supporting real-time monitoring through an IoT application. This overall architecture is illustrated in the block diagram shown on **page 4**.

Features of the Project

1. Helmet Detection

Problem

Many riders travel without wearing helmets.

Solution

An **IR sensor** is placed inside the helmet.

When the rider wears the helmet:

- IR sensor detects the rider's head.
- Helmet status is verified.
- Bike ignition can proceed.

If helmet is not worn:

- Bike cannot start.

Benefit

- Makes helmet usage compulsory.
- Reduces head injuries.

(Page 5 shows the IR sensor used for helmet detection.)

2. Alcohol Detection

Problem

Drunk driving is one of the leading causes of road accidents.

Solution

An **MQ-3 Alcohol Sensor** checks the rider's breath.

If alcohol is detected:

- Ignition is disabled.
- Rider is prevented from starting the bike.

Benefit

- Prevents drunk driving.
- Improves road safety.

(The MQ-3 alcohol sensor is described and illustrated on **page 6**.)

3. Accident Detection

Problem

After an accident, valuable time is lost before anyone knows the rider needs help.

Solution

An **accelerometer** continuously measures sudden movement, impact, or free fall.

If abnormal motion is detected:

- Accident is identified automatically.
- Emergency process begins immediately.

Benefit

- Faster medical response.
- Reduces accident fatalities.

(The accelerometer module is described on **page 7**.)

4. GPS Location Tracking

Problem

Emergency responders often do not know the rider's exact location.

Solution

A **NEO-6M GPS Module** continuously tracks the bike.

If an accident occurs:

- GPS obtains precise latitude and longitude.
- Location is sent with the emergency alert.

Benefit

- Ambulance reaches the rider faster.
- Family receives accurate accident location.

(The GPS module is explained on **page 7**.)

5. GSM Emergency Alerts

Problem

Family members and emergency services are informed too late.

Solution

A **SIM900 GSM Module** automatically sends:

- SMS
- Emergency notification
- Rider's GPS location

to predefined contacts.

Benefit

- Immediate emergency communication.
- Faster rescue operations.

(The GSM module is discussed on **page 8**.)

6. IoT Real-Time Monitoring

The project also supports **real-time monitoring** using an IoT application.

Users can:

- Monitor rider status
- View alerts
- Track emergency events

Screenshots of the monitoring interface are shown on **page 12**.

Benefit

- Continuous monitoring.
 - Better fleet and rider management.
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7. Automatic Bike Ignition Control

The system checks:

☒ Helmet worn

AND

✓ Alcohol not detected

Only then:

- Relay activates.
- Bike ignition starts.

Otherwise:

- Bike remains OFF.

(The relay module used for ignition control is shown on **page 9**.)

8. LCD Display

The helmet system includes a **16×2 LCD**.

It displays:

- Helmet status
- Alcohol detection
- Accident alerts
- System messages

This helps the rider understand the system status easily.

(The LCD module and sample outputs appear on **pages 9–12**.)

Components Used

Component	Purpose
Arduino UNO	Main microcontroller
IR Sensor	Detects helmet usage

MQ-3 Alcohol Sensor	Detects alcohol consumption
Accelerometer	Accident detection
GPS Module (NEO-6M)	Rider location tracking
GSM Module (SIM900)	Sends SMS and emergency alerts
Relay Module	Controls bike ignition
LCD Display (16×2)	Displays system status
IoT Mobile Application	Real-time monitoring

These hardware modules are introduced and illustrated throughout **pages 4–9** of the paper.

Working of the System

Step 1

Rider wears the helmet.

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Step 2

IR sensor checks whether the helmet is worn.

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Step 3

MQ-3 sensor checks alcohol level.



Step 4

Arduino UNO processes all sensor inputs.



Step 5

If helmet is worn and alcohol is not detected:

- Relay turns ON.
- Bike ignition starts.



Step 6

During riding, the accelerometer continuously monitors motion.



Step 7

If an accident is detected:

- GPS determines the rider's location.
- GSM sends SMS with location.
- IoT application receives an alert.
- LCD displays the accident status.

The overall flow is illustrated in the system block diagram on **page 4**, with integrated hardware shown on **page 10** and example outputs on **pages 11–12**.

Advantages

- Makes helmet usage compulsory.

- Prevents drunk driving.
- Automatically detects accidents.
- Sends emergency alerts with GPS location.
- Supports real-time IoT monitoring.
- Provides live rider tracking.
- Low-cost IoT safety solution.
- Faster emergency response.
- Easy to integrate with motorcycles.

These are consistent with the objectives and conclusion presented in the paper.

Limitations

While the paper highlights the benefits, some practical limitations are implied:

- Requires additional hardware installation on the bike and helmet.
- Depends on GSM network availability for sending alerts.
- GPS accuracy may be affected in tunnels or areas with poor satellite visibility.
- Sensors require proper calibration for reliable detection.
- Battery maintenance is necessary for continuous operation.

The paper itself does not provide a separate "Limitations" section; these are practical considerations inferred from the described design.

Future Scope

The paper concludes that the system can be enhanced through continued research and development. Possible future improvements include:

- More advanced IoT integration.
- Smarter accident detection algorithms.
- Improved real-time monitoring.
- Better communication technologies.
- Additional safety sensors.
- Enhanced mobile application features.
- AI-based rider safety analysis.

The conclusion emphasizes continued development to create even more advanced smart helmets.

Final Summary

This project presents an **IoT-based Smart Helmet System** that combines **helmet detection, alcohol detection, accident detection, GPS tracking, GSM emergency messaging, automatic bike ignition control, LCD status display, and real-time IoT monitoring** into a single safety solution. The system prevents the motorcycle from starting unless the rider is wearing a helmet and has not consumed alcohol. During the ride, it continuously monitors for accidents and, if one occurs, automatically sends the rider's location to emergency contacts while displaying alerts locally and through an IoT application. Overall, the project aims to reduce road accidents, improve rider safety, and enable faster emergency response using low-cost IoT technologies.